

PHY201: Waves & Optics

Problem Set 1

Aug 10, 2025

1. A mass of 0.75 kg is attached to one end of a horizontal spring of spring constant 400 N-m⁻¹. The other end of the spring is attached to a rigid wall. The mass is pushed so that at time $t = 0$ it is 4.0 cm closer to the wall than the equilibrium position and is travelling towards the wall with a velocity of 0.50 m s⁻¹. (a) Determine the total energy of the oscillating system. (b) Obtain an expression for the displacement of the mass in the form $x = A\cos(\omega t + \phi)$ m, giving numerical values for A , ω and ϕ .
2. A uniform rod of length L is nailed to a post so that two-thirds of its length is below the nail. What is the period of small oscillations of the rod?
3. A wire of unstretched length l_0 is extended by a distance 10^{-3} when a certain mass is hung from its bottom end. If this same wire is connected between two points, A and B, that are a distance l_0 apart on the same horizontal level, and the same mass is hung from the mid-point of the wire, what is the depression y of the midpoint, and what is the tension in the wire?
4. A mass M oscillates at the end of a spring that has spring constant k and finite mass m . (a) Show that the total energy E of the system for oscillations of small amplitude is given by

$$E = \frac{1}{2}(M + m/3)v^2 + \frac{1}{2}kx^2$$

where v and x are the velocity and displacement of the mass M , respectively. (b) Hence, show that the frequency of the oscillations is equal to $\sqrt{k/(M + m/3)}$.

5. A mass m hangs from a uniform spring of spring constant k . (a) What is the period of oscillations in the system? (b) What would it be if the mass m were hung so that (1) it was attached to two identical springs hanging side by side? (2) It was attached to the lower of two identical springs connected end to end?
6. (a) A solid steel ball is to be hung at the bottom end of a steel wire of length 2 m and radius 1 mm. The ultimate strength of steel is 1.1×10^9 N/m²?. What are the radius and the mass of the biggest ball that the wire can bear? (b) What is the period of torsional oscillation of this system? (Shear modulus of steel = 8×10^{10} N/m². Moment of inertia of sphere about axis through centre = $2MR^2/5$.)